PIERNIK · BRANCH SF_STREAMINGCR_COMB · UNCOMMITTED WORKING TREE

# Unsplit solver, positivity safeguards and energy-consistent CR coupling: change log

Base commit **95e7eb907** · Date **13 October 2026** · Prepared for **Vinod V. Pisharody**

All edits made while moving the five tallbox_sf comparison cases (A HD, B MHD, C1 diffusive CR, C2 two-moment CR diffusion, D streaming CR) to the unsplit Riemann solver, and while tracking down the three crashes that followed. For each change the record gives the code before and after and the reason. Section B describes how each crash was traced, section C the validation, section D what was deliberately left unchanged. Line numbers refer to the current files.

**11**
records: 7 source files, 4 problem/analysis files

**4**
bugs fixed that gave NaN, missing physics or non-conservation

**3**
crashes resolved: A at 0.5 Myr, D at 25.17 Myr, C2 at 96.996 Myr

**1680 → 0**
faces/step needing the positivity fallback in the C2 crash window, after the CR fix

BUG FIX wrong results, crash or NaN

SAFEGUARD acts only on unphysical states

DIAGNOSTIC measures, does not change results

CONFIG parameters, documentation, analysis

## Contents


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Then: **B** how the crashes were traced · **C** validation · **D** open issues and unchanged limiters · **E** state of the runs.

## A. Overview: before, after, motivation

| # | BEFORE | AFTER | MOTIVATION |
|---|---|---|---|
| 1 | No star formation with the unsplit solver | `SF_step` called in the unsplit sweep | C2/D need the unsplit solver |
| 2 | Floors only after the last RK stage | Floors after every stage, as in the split solver | Case A NaN at step 2 |
| 3 | Negative face pressure went into HLLD | First-order fallback, floor only if the cell itself is non-positive | C2/D NaN in the halo |
| 4 | Ghost cells: E copied, B extrapolated | Ghost E rebuilt from edge pressure | $p < 0$ in every z-boundary ghost cell |
| 5 | $-dt\nabla P_c$ kick without energy; guard created CR energy | Momentum $-\Delta F_c/c^2$, conserved energy, capped exchange | Root cause of the remaining $p < 0$ cells; energy conservation |
| 6–8 | Floor energy and safeguard activity unknown | Four new tsl columns | Show the safeguards are rare and quantify them |
| 9–11 | Split solver; whole-box phase plot only | Unsplit solver, `cfl_glm`; wind and flux plots | Configuration and wind analysis |

## Records

### 1 Star formation in the unsplit solver

BUG FIX

`src/scheme/unsplit_schemes/unsplit_fluidupdate.F90` · lines 101 and 183-185

**BEFORE**

```
      call external_sources(forward)
      if (associated(problem_customize_solution)) call problem_customize_solution(forward)

      call eglm
      call glmdamping
```

**AFTER**

```
#ifdef NBODY
      use star_formation,      only: SF_step
#endif /* NBODY */
      ...
      call external_sources(forward)
      if (associated(problem_customize_solution)) call problem_customize_solution(forward)
#ifdef NBODY
      call SF_step(forward)
#endif /* NBODY */
      call eglm
      call glmdamping
```

**WHY** `SF_step` was called only from the split sweep (`make_3sweeps`). The unsplit sweep `make_unsplitsweep` is a copy of it that never received the call, so with `solver_str = 'riemann_unsplit'` no stars formed and no SN feedback ran, silently. The streaming-CR module is built for the unsplit solver, so cases C2 and D need it. The call is placed exactly as in the split sweep; the unsplit driver also calls the sweep with `forward = .true.` then `.false.`, which is all `SF_step` relies on.

## 2 Floors after every Runge–Kutta stage

BUG FIX

`src/scheme/unsplit_schemes/unsplit_source.F90` · lines 106–116

**BEFORE**

```
call internal_sources(size(u, 1, kind=4), u, u1, b, cg, istep, ddim, i1, i2, rk_coef(istep) * dt, vx)

if (istep == last_stage(integration_order)) then
   call care_for_positives(size(u, 1, kind=4), u1, b, cg, ddim, i1, i2)
endif
```

**AFTER**

```
call internal_sources(size(u, 1, kind=4), u, u1, b, cg, istep, ddim, i1, i2, rk_coef(istep) * dt, vx)

! every stage, as in the split solver: a negative pressure in the intermediate state gives NaN face
states in the next stage
call care_for_positives(size(u, 1, kind=4), u1, b, cg, ddim, i1, i2)
```

**WHY** Case A (HD) went NaN at step 2 and lost almost all its mass within 2 Myr. Instrumenting the face states showed that every bad face occurred in the second RK stage, reconstructed from intermediate-state cells with negative pressure next to a fresh SN (kinetic-energy dominated gas). `riemann_hlld_u` takes `sqrt(γp/ρ)`, so p < 0 gives NaN. The split solver applies the density/internal-energy floors after every stage (`solve_cg_riemann.F90`, two calls); the unsplit one applied them only after the last stage. Cases B–D survived only because the magnetic terms kept the HLLD wave-speed argument positive. Same change in the non-magnetic branch.

## 3 Positivity fallback for face states

SAFEGUARD

`src/scheme/Riemann_Solver/interpolations.F90` · line 139 (call), lines 141–196 (new routine), counters at lines 42–45

**BEFORE**

```
      q = utoq(u, bcc)
      call interp(q,   ql,   qr,   flimiter)
      if (present(bcc)) call interp(bcc, bccl, bccr, blimiter)

   end subroutine interpol
```

**AFTER**

```
      q = utoq(u, bcc)
      call interp(q,   ql,   qr,   flimiter)
      if (present(bcc)) call interp(bcc, bccl, bccr, blimiter)

      call positive_face_states(q, ql, qr, bcc, bccl, bccr)

   end subroutine interpol

   subroutine positive_face_states(q, ql, qr, bcc, bccl, bccr)
      ...
      bad = (ql(i, fl%idn) <= 0.) .or. (qr(i, fl%idn) <= 0.)
      if (fl%has_energy) bad = bad .or. (ql(i, fl%ien) <= 0.) .or. (qr(i, fl%ien) <= 0.)
      if (.not. bad) cycle
      ql(i, fl%beg:fl%end) = q(i,   fl%beg:fl%end)      ! first-order states, fluid ...
      qr(i, fl%beg:fl%end) = q(i+1, fl%beg:fl%end)
      bccl(i, :) = bcc(i, :) ; bccr(i, :) = bcc(i+1, :)  ! ... and magnetic field together
      ql(i, fl%idn) = max(ql(i, fl%idn), smalld)        ! only if the cell itself is non-positive
      ql(i, fl%ien) = max(ql(i, fl%ien), smallp)
      ...                                               ! counted in n_face_fallback / n_face_floor
```

**WHY**

C2 crashed at t = 96.996 Myr (D at 25.17 Myr). Replaying from a restart with NaN checks after each operation located the first non-finite value in the MHD sweep and then the first non-finite HLLD flux: an interior halo face with $\rho \approx 10^{-5}$, v ≈ 75 pc/Myr and both reconstructed pressures negative ($-8.7\times10^{-5}$, $-1.7\times10^{-4}$). The standard remedy, and the same idea as `scr_fallback_1st` already used for the CR solver, is to use first-order states on such faces; the update stays conservative because fluxes are still differenced. Physical states are not touched.

This was first added as the crash fix. The investigation then traced the non-positive cells to two root causes, fixed in records 4 and 5; with those in place the fallback acts on about 1–2 faces per step out of ~$1.3\times10^{6}$ (section C).

## 4 Thermodynamically consistent outflow ghost cells

BUG FIX

`src/grid/cg_list_bnd.F90` · calls at lines 870 (BND_OUT) and 892 (BND_OUTD); new routine outflow_energy, lines 1001-1077

**BEFORE**

```
case (BND_OUTD)
   ! all of cg%u (rho, rho*v, E) copied from the edge cell into every ghost layer
   cg%u(:, ghost) = cg%u(:, edge)
   ! diode: outward normal momentum enforced in the ghosts, E unchanged
   ...
! magnetic ghosts, set separately in outflow_b:
!   B_normal(ghost) = 2 B_normal(next inward) - B_normal(next-but-one)   (linear extrapolation)
!   B_tangential(ghost) = B_tangential(edge)                              (copy)
```

**AFTER**

```
case (BND_OUTD)
   ... as before (copy, diode) ...
   call outflow_energy(cg, dir, side)

subroutine outflow_energy(cg, dir, side)
   ! p_edge = (gamma-1) (E - rho v^2/2 - B^2/2) of the edge cell, floored at smallp
   ! for every ghost layer, with the ghost rho, v (after the diode) and the ghost B
   ! evaluated with the same recursion and operation order as outflow_b:
   !    E_ghost = p_edge/(gamma-1) + rho v^2/2 + B^2/2
```

**WHY**

Gas pressure is not stored; it is derived as $p = (\gamma-1)(E - \rho v^2/2 - B^2/2)$. Copying E while $B_{normal}$ is extrapolated (and the momentum is changed by the diode) pairs the edge-cell E with a different B and v. In the magnetically dominated halo this made the ghost pressure negative on every step along the whole z boundary. Example from the debug run: $B_{normal}$ −0.252 (edge) vs −0.269 (ghost), $\Delta(B^2/2) = 4.3\times10^{-3}$ against a thermal energy of $1.4\times10^{-6}$, so p went from $+9.3\times10^{-7}$ to $-2.86\times10^{-3}$. These faces were about 80% of the fallback events of record 3 at the time. Rebuilding E from the edge pressure gives the zero pressure gradient an outflow boundary is meant to impose. Placed in the fluid boundary routine because many code paths refill fluid ghosts without refilling B; the ghost B is recomputed from active cells only, so the result does not depend on the order of fluid and magnetic boundary updates. Only the ghost cells at |z| = 2 kpc change; they set the fluxes leaving the box, so mass and energy outflow through the boundary become more consistent.

## 5 Energy-consistent CR–gas coupling

BUG FIX

src/fluids/cosmicrays_streaming/streamingcr_source.F90 · lines 135, 178–204 and 230–234

**BEFORE**

```
! inside the cell loop (energy only)
if (.not. disable_feedback ) then  ! Energy feedback to the MHD gas . Only the first fluid
   e_feed = cg%w(uhi)%arr(iarr_all_en(1), i, j, k) - (newec - ec)
   if ((e_feed < 0) .or. (newec .equals. smallescr)) then
      e_feed = cg%w(uhi)%arr(iarr_all_en(1), i, j, k)      ! gas unchanged, newec kept
   endif
   cg%w(uhi)%arr(iarr_all_en(1), i, j, k) = e_feed
endif

! after the loop (momentum only)
cg%w(gpci)%arr(:,:,:,:) = cg%get_gradient(ord = ord_pc_grad, iw = scri, vec = iarr_all_escr)
cg%w(uhi)%arr(iarr_all_mx(1), :,:,:) = cg%w(uhi)%arr(iarr_all_mx(1), :,:,:) &
     - 1.0/3.0 * dt_scr * rk_coef(istep) * cg%w(gpci)%arr(xdim + 3 * (ns - 1),:,:,:)
...                                                          ! same for y, z
```

**AFTER**

```
f1l = f1 ; f2l = f2 ; f3l = f3                     ! lab-frame Fc/cred before the source step
...
if (.not. disable_feedback) then
   ! (1/c^2) dFc/dt = -sigma (Fc - 4/3 Ec v),  d(rho v)/dt = +sigma (Fc - 4/3 Ec v)
   ! gas gets the momentum the CR flux loses: Delta(rho v) = -Delta(Fc)/cred^2   (Jiang & Oh 2018)
   cg%w(uhi)%arr(iarr_all_mx(1), i, j, k) = cg%w(uhi)%arr(iarr_all_mx(1), i, j, k) - (newf1 - f1l) / cred
   ...                                                 ! y, z
   ! total (gas + CR) energy conserved; CR gain capped by the available gas internal energy
   e_avail = E_gas - E_mag - smallei - E_kin(after the momentum update)
   de_c = newec - ec
   if (de_c > max(e_avail, 0.0)) then
      de_c = max(e_avail, 0.0) ; newec = ec + de_c
      scr_nexch_cap = scr_nexch_cap + 1
   endif
   E_gas = E_gas - de_c
endif

! after the loop: grad(Pc) still refreshed for later use, no momentum kick
cg%w(gpci)%arr(:,:,:,:) = cg%get_gradient(ord = ord_pc_grad, iw = scri, vec = iarr_all_escr)
```

**WHY**

With the fixes above, the remaining non-positive cells were counted after each operation of one step: about 350 cells per half-step, all created by the streaming-CR sweep and none by anything else. Two inconsistencies:

(1) The gas energy exchange used the implicit solution ($-\Delta E_c$), but the momentum was given by a separate $-dt\nabla P_c$ kick with no matching energy change. The kick's kinetic energy, including the $\Delta m^2/2\rho$ term, came out of the internal energy; at $P_{CR}/P_{gas} \sim 10^6$ and $\rho \sim 10^{-5}$ it exceeded it. The two forms agree only once the flux has relaxed to equilibrium.

(2) The guard tested the *total* gas energy, and when it fired it skipped the gas update but kept the new $E_c$, so the CRs gained energy from nothing.

Now momentum and energy come from the same implicit solution, total energy is conserved exactly, and the CR energy gain cannot exceed the gas internal energy (a conservative version of an energy floor, counted).

## 6 Floor energy accounting

DIAGNOSTIC

`src/fluids/sources.F90` · lines 36–38 and 310–330

**BEFORE**

```
call limit_minimal_density(n, u1, cg, sweep, i1, i2)
call limit_minimal_intener(n, bb, u1)
```

**AFTER**

```
call limit_minimal_density(n, u1, cg, sweep, i1, i2)
e_before = sum(u1(dom%nb+1:n-dom%nb, iarr_all_en))          ! active cells only
call limit_minimal_intener(n, bb, u1)
floor_energy = floor_energy + (sum(u1(dom%nb+1:n-dom%nb, iarr_all_en)) - e_before) * cg%dvol
```

**WHY** The density floor was already tracked (`ion_mmass_cum`), but the energy added by the internal-energy and 10 K temperature floors was not. Since record 2 applies these floors more often, the energy they add is now measured and written to the tsl (record 8). Results are unchanged.

## 7 Counter for capped CR exchange

DIAGNOSTIC

`src/fluids/cosmicrays_streaming/initstreamingcr.F90` · line 97

**BEFORE**

```
(no counter)
```

**AFTER**

```
integer :: scr_nexch_cap  = 0   ! cells where the CR energy gain was capped by the available gas internal
energy
```

**WHY** Counts how often the cap of record 5 acts, alongside the existing streaming-CR counters.

## 8 New tsl columns

DIAGNOSTIC

`problems/tallbox_sf/initproblem.F90` · lines 692–693, 720–734, 785–797, 814

**BEFORE**

```
... "npp_limited   ", "nfloor        "])
```

**AFTER**

```
... "npp_limited   ", "nfloor        ", "nexch_capped  "])        ! STREAM_CR builds
"n_face_fb     ", "n_face_floor  ", "E_floor       "              ! all builds, appended at the end
```

**WHY** All four are summed over ranks and reset at every tsl dump (per interval, like the other safeguard counters): `n_face_fb` faces next to active cells reduced to first order (summed over sweeps and RK stages); `n_face_floor` those among them that also needed a floor; `E_floor` energy added by the floors [code units]; `nexch_capped` cells with a capped CR exchange. The new columns are appended so existing column positions do not move.

## 9 Unsplit solver and GLM CFL

CONFIG

`problems/tallbox_sf/problem.par, .A, .B, .C1, .C2, .D · NUMERICAL_SETUP, lines 46-48`

**BEFORE**

```
solver_str = 'riemann'      ! split solver: star formation (SF_step) is not called by the unsplit solver
cfl    = 0.3
```

**AFTER**

```
solver_str = 'riemann_unsplit'   ! unsplit MHD (needs divB_0 = 'HDC', the default, and cfl <= 0.3); ...
cfl    = 0.3
cfl_glm = 0.3               ! GLM cleaning-wave CFL; defaults to the built-in cfl, not the one above, ...
```

**WHY** All five cases now use the unsplit solver (record 1 made this possible). `cfl_glm` must be set explicitly: `global.F90` sets `cfl_glm = cfl` before the namelist is read, so it inherits the built-in default 0.7, not the 0.3 given in the parameter file, and the unsplit MHD solver warns above 0.3.

## 10 Wind diagnostics and tsl reader

CONFIG

`problems/tallbox_sf/analysis/compare.py · read_tsl; new read_kappa, outward, cr_flux_z, plot_wind_phase, plot_fluxes`

**BEFORE**

```
# read_tsl: column names from the first file only; files with a different
#           number of columns could not be combined
# plots: timeseries, profiles, slices, mass-weighted phase diagram (whole box)
```

**AFTER**

```
# read_tsl: each file read with its own header; union of columns, NaN where missing
# wind_phase_<t>.png : n-T diagram of outflowing gas only (|z| > 500 pc, v_z away from the
#                      midplane), weighted by mass flux; hot (T > 2e4 K) share of the flux
# fluxes_<t>.png     : vs |z| (both sides summed): net outward Mdot, Mdot of outflowing hot
#                      gas, outward CR energy flux (C2/D: evolved Fc_z; C1: 4/3 e v_z minus
#                      anisotropic diffusion with kappa from problem.par)
```

**WHY** The whole-box phase diagram cannot tell outflowing gas from hot gas inside SN bubbles, so the wind comparison needed diagnostics restricted to outflowing gas and vertical flux profiles. Check: the snapshot mass flux through |z| = 500 pc reproduces the tsl `Mdot_z1` to all printed digits in all five runs. The reader change is needed because the new tsl columns give old and new tsl files different layouts.

## 11 Documentation

CONFIG

`problems/tallbox_sf/info · solver paragraph`

**BEFORE**

```
Star formation needs a split solver (solver_str = 'riemann'); the unsplit solver does not call SF_step.
```

**AFTER**

```
All variants use the unsplit Riemann solver (solver_str = 'riemann_unsplit', GLM divergence cleaning, cfl
= 0.3);
SF_step is called from both the split (make_3sweeps) and the unsplit (make_unsplitsweep) sweeps, ...
The unsplit solver applies the density/internal-energy floors (care_for_positives) after every RK stage,
...
cfl_glm must be set explicitly (0.3): its default is the built-in cfl, not the cfl given in the parameter
file.
```

**WHY** Keeps the problem notes in line with records 1, 2 and 9.

## B. How the crashes were traced

Each crash was replayed deterministically from a restart file shortly before it, with temporary instrumentation (removed afterwards; no debug code remains in the tree).

| CRASH | SIGNATURE | LOCATED BY | CAUSE | FIX |
|---|---|---|---|---|
| A (HD), step 2 | NaN momentum from step 2, dt doubling every step, gas mass $1.06\times10^{7}$ → $4\times10^{3}$ $M_\odot$ | face-state check in `solve_u`: all bad faces in RK stage 2 (`10002`), cell p < 0 in the intermediate state | floors skipped after stage 1 | record 2 |
| C2, t = 96.996 Myr | NaN over z = 300–1200 pc, then 17 NaN restarts while running on | NaN check after each operation → MHD sweep; flux check → first non-finite HLLD flux at a face with both $p_L$, $p_R$ < 0 | non-positive pressure reaching HLLD | record 3 (then 4, 5) |
| D, t = 25.17 Myr | same dt-doubling signature | same mechanism (crashed run had no nearby restart) | as C2 | records 3–5; D passed 25.17 Myr afterwards |

### Where the non-positive cells came from

| STEP OF THE INVESTIGATION | FACES/STEP AT P ≤ 0 | FINDING |
|---|---|---|
| C2 at 96.8 Myr, fallback only (debug run) | ~900 | about 80% on z-boundary ghost faces → record 4 |
| after record 4, locating the remaining ones | ~1700 | all with a non-positive *cell* pressure, in block interiors |
| count of p ≤ 0 cells after each operation | — | ~350 cells per half-step created by the streaming-CR sweep, 0 by anything else → record 5 |
| after record 5 (same window) | 0 | fallback no longer acts; ~100 capped exchanges/step instead |

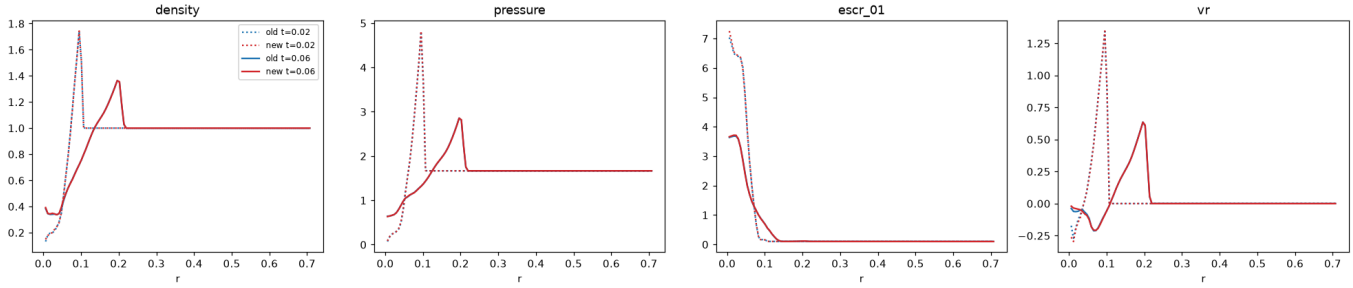
## C. Validation

### Streaming-CR test problems (problems/scr_test), old vs new coupling (record 5)

| TEST | QUANTITY | OLD | NEW |
|---|---|---|---|
| 7: CR-driven waves, 1D periodic, σ = 0.5 | total energy (gas + CR) at t = 0.05, initial 87.999999911 | 87.999999866 | 87.999999925 |
| | max difference new−old in ρ, p, $E_c$ / in $v_x$ | ≤ $1\times10^{-3}$ / $2.5\times10^{-2}$ of max | |
| | minimum gas pressure | 0.984 | 0.985 |
| 8: CR-driven blast wave, 2D 512², σ = $10^6$ | L1 relative difference at t = 0.06: ρ / p / $E_c$ / \|v\| | $7\times10^{-4}$ / $9\times10^{-4}$ / $5\times10^{-3}$ / $3\times10^{-2}$ | |
| | total energy at t = 0.06 (initial 3.2265213) | 3.2247111 | 3.2247027 |
| | minimum gas pressure | 0.429 | 0.441 |

Test 7 has weak coupling, so the flux is far from equilibrium and the old $\nabla P_c$ kick and the actual coupling force differ; the 2.5% velocity difference is expected there. In test 8 (strong coupling) the two must agree, and they do:

the shock positions and profiles coincide (figure). The larger max-norm differences are confined to the innermost cells, where v ≈ 0. The energy drift in test 8 is the same for old and new (outflow boundaries).




Test 8: azimuthally averaged radial profiles of density, pressure, CR energy and radial velocity at t = 0.02 (dotted) and 0.06 (solid); old coupling blue, new red.

### C2 crash window (96.80 → 97.00 Myr, 44 steps, identical dt)

| PER STEP | OLD COUPLING | NEW COUPLING |
|---|---|---|
| faces at first order (`n_face_fb`) | 1680 | 0 |
| faces needing a floor (`n_face_floor`) | 1680 | 0 |
| capped CR exchanges (`nexch_capped`) | — | 102 |
| CR flux clips (`nclip_src`) | 1.0 | 1.5 |
| NaN | none | none |

### Production runs, first ~850 steps after switching to the new coupling

| PER STEP | C2 BEFORE | C2 AFTER | D BEFORE | D AFTER |
|---|---|---|---|---|
| faces at first order | 107 | 1.0 | 13 | 1.7 |
| capped CR exchanges (of ~$4.4\times10^5$ cells) | — | 12.5 | — | 30 |
| floor energy per Myr [code] | $2.1\times10^7$ | $1.3\times10^7$ | $1.7\times10^7$ | $2.5\times10^7$ |

Other checks: case A runs to 25 Myr with gas mass conserved ($1.05592\times10^7$ vs $1.05590\times10^7$ at t = 0 after 3 Myr); cumulative density-floor mass is $1.3–3.7\times10^{-4}$ of the gas mass in every run; C2 and D have run past their old crash times (C2 to > 229 Myr, D to > 142 Myr) without NaN. All builds compile without warnings in the changed files, including the HD build (`NONMAGNETIC` paths).

## D. Open issues and what was left unchanged

### Remaining floor energy (accepted, monitored)

The floors still add $1.3–2.5\times10^7$ code units per Myr in C2/D, about 1% of the SN energy input (1 SN = $10^{51}$ erg ≈ $5.3\times10^7$ code units; ~40 SNe/Myr). A classified run attributed about 97% of it to ~290 cells per step whose internal energy goes negative inside the MHD sweep, at $n_H \approx 0.03–0.3$ cm$^{-3}$; the 10 K temperature floor contributes ~3%. This is the standard high-Mach-number error of total-energy schemes in kinetic-energy dominated, SN-driven flow, not the CR coupling. Options if it ever matters: check whether these cells cluster at recent SN sites (pure kinetic injection with the 2000 pc/Myr cap), or a dual-energy formalism. Decision: accepted and monitored through `E_floor`.

### To be checked

> **CR boundary in C2/D (suspected, not verified).** At 140 Myr $P_{CR}$ is flat to the z boundary (~4000 K cm$^{-3}$) in C2/D, while in C1 it drops to zero (`diff_bnd_str = "zero"`). $E_{CR}/E_{th}$ stays at 0.3–0.8 in C1 but rises to 3–10 (D) and up to ~30 (C2). A zero-gradient CR boundary in the two-moment runs would let CR energy build up for numerical reasons and make the CR budget of C1 and C2/D not comparable. The streaming-CR boundary code has not been checked yet.

### Pre-existing limiters, not changed

| LIMITER | WHERE | EFFECT |
|---|---|---|
| SN kick velocity cap 2000 pc/Myr | `TIGRESS_injection` | removes momentum/energy from SNe in very low density |
| Alfvén-speed ceiling `va_max` | `alfven_limiter.F90` | adds mass; 0–7 $M_\odot$ so far |
| CR flux causality clip, positivity limiter, 1st-order fallback | streaming-CR transport/source | counted in the tsl (`nclip_*`, `npp_limited`, `nfallback_1st`) |
| $E_c$ floor `smallescr` | `apply_scr_source` | applied after the exchange; adds CR energy where $E_c$ would fall below it |
| 10 K temperature floor | `limit_minimal_intener` (THERM) | ~3% of `E_floor` |
| External boundaries not reapplied to the intermediate RK array | `bnd_u`, `bnd_b` act on `cg%u`, `cg%b` | first-order in time at the domain boundary |

## E. State of the runs

| RUN | CODE HISTORY | REACHED |
|---|---|---|
| A | records 1–2; stopped as planned | 25 Myr |
| B, C1 | records 1–2 only (built before records 3–8); paused with restarts | 163.3 / 145.4 Myr |
| C2 | old CR coupling to 118.08 Myr (record 3 from 95.0, record 4 and counters from 115.16), all records after | > 229 Myr |
| D | 0–9.7 Myr original run, rerun from 9.7 (record 3 from 15.09, record 4 and counters from 28.93), all records from 31.68 Myr | > 142 Myr |

Each C2/D run directory has a `CODE_CHANGES.txt` with the switch times. All runs show a large start-up oscillation (field-SN driving stops at 40 Myr; disks swell to ~600 pc, fall back, swell again by 140 Myr), so comparisons before ~150 Myr reflect the start-up phase. For a clean comparison all five cases should be rebuilt with the current tree and rerun from t = 0.